# Beyond Play Count: Exploring Advanced Time Series Analyses and "Listening Vibes" for Music Data

An In-Depth Exploration of Possibilities, Metrics, and Machine Learning Approaches Using `fm_data.csv`

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March 22, 2026

**Abstract**

The initial project report established a baseline approach: aggregating Last.fm scrobble data into daily play counts and forecasting them with SARIMA. While effective as a starting point, raw play count is the *least informative* metric we can extract from the data. This document explores a rich landscape of alternative analyses — from defining temporal "listening vibes" (clusters of songs based on when they were played) to computing diversity indices, discovery rates, loyalty scores, and session dynamics. For each possibility, we discuss what the metric captures, how it would be constructed from `fm_data.csv`, what time series plots it enables, and which machine learning models are most suitable for forecasting or classification. No code is included; the focus is purely on conceptual exploration and modelling strategy.

## Contents

# 1 Motivation: Why Move Beyond Play Count?

Daily play count answers a single question: *How much did the user listen?* But it says nothing about:

- **When** they listened (morning vs. late night),
- **What** they listened to (one artist on repeat vs. dozens),
- **How** they listened (short background sessions vs. long focused ones),
- **Whether** they were exploring new music or revisiting favourites.

Our dataset contains **Username, Artist, Track, Album, Date, and Time** for 166,153 scrobbles across 11 users over 31 days. This is far richer than a single count. The following sections explore every meaningful metric and analysis we can extract from it.

# 2 The "Listening Vibes" Framework

## 2.1 What Are Vibes?

A **vibe** is a temporal cluster — a named time window defined by when a scrobble occurred. The hypothesis is simple: *people listen to different kinds of music at different times, and these temporal slots carry distinct emotional and behavioural signatures.*

Rather than treating time as a raw number (hour 0–23), we assign each scrobble to a meaningful, human-interpretable category.

## 2.2 Proposed Vibe Definitions

We propose two tiers of vibes: **primary** (based on time of day) and **compound** (combining day of week with time of day).

### 2.2.1 Primary Vibes (Time-of-Day)

| Vibe | Hours | Behavioural Hypothesis |
|---|---|---|
| Early Morning | 05:00–08:59 | Wake-up routines, commute music. Likely energetic or habitual. |
| Morning | 09:00–11:59 | Focus and productivity. Instrumental, ambient, or familiar. |
| Afternoon | 12:00–16:59 | Midday energy. Mixed genres, background listening. |
| Evening | 17:00–20:59 | Wind-down or social. More intentional, curated listening. |
| Late Night | 21:00–00:59 | Reflective, moody, or intense. Deeper engagement. |
| Deep Night | 01:00–04:59 | Insomnia or marathon sessions. Niche, experimental. |

### 2.2.2 Compound Vibes (Day + Time)

| Vibe | Definition | Behavioural Hypothesis |
|---|---|---|
| Weekend Vibes | Sat–Sun, any time | Leisure listening, longer sessions, more exploration. |
| Weekday Grind | Mon–Fri, 09–17 | Work-time music, focus-oriented, lower diversity. |
| Friday Night | Fri 18:00 – Sat 03:00 | Social, party, high-energy or celebratory. |
| Sunday Chill | Sun, all day | Recovery, mellow, singer-songwriter, acoustic. |
| Monday Morning | Mon, 05:00–11:59 | Re-entry to the work week, motivational or comforting. |

## 2.3 What Makes Vibes Analytically Powerful?

Vibes are not arbitrary labels — they are **categorical features** that enable a fundamentally new type of analysis. Once every scrobble is tagged with a vibe, we can:

1. **Compare distributions**: Which artists or tracks dominate each vibe? Do Late Night and Morning vibes share any overlap?
2. **Track vibe proportions over time**: Plot the percentage of each vibe per week as a stacked area chart. This reveals behavioural shifts (e.g. a user gradually becoming a night listener).
3. **Cross vibes with other metrics**: Compute diversity, discovery rate, or loyalty score *within each vibe separately*. This is far more revealing than global metrics.
4. **Use vibes as features in ML models**: Pass vibe proportions as exogenous variables to forecasting models.

## 2.4 Vibes as a Time Series

### 2.4.1 Vibe Composition Plot (Stacked Area)

For each time unit (day or week), compute the proportion of scrobbles falling into each vibe. Plotted as a **stacked area chart**, this shows how the user's daily rhythm evolves over the 31-day window.

**What to look for**:

- Do weekends consistently show a different vibe mix?
- Are there days where Late Night and Deep Night vibes spike (suggesting binge sessions)?
- Does the vibe composition remain stable, or does it drift over the month?

### 2.4.2 Vibe Volume Plot (Absolute Counts)

Instead of proportions, plot the raw number of scrobbles per vibe per day. This combines the "how much" of play count with the "when" of vibes, giving a much richer picture.

#### 2.4.3 Machine Learning Approach for Vibes

**Forecasting vibe proportions.** Each day produces a vector of vibe proportions $\mathbf{v}_t = (v_t^{\text{morning}}, v_t^{\text{afternoon}}, \ldots)$. This is a **compositional time series** (proportions that sum to 1).

- **Vector Autoregression (VAR)**: Treats each vibe proportion as one variable in a multivariate time series. VAR captures cross-correlations — e.g. a spike in Late Night today might predict a drop in Early Morning tomorrow.
- **Dirichlet Regression**: Since vibes are compositional (proportions summing to 1), Dirichlet regression is the theoretically correct model. It respects the simplex constraint and avoids the problems of modelling proportions independently.
- **LSTM / GRU**: A recurrent neural network that takes the vibe vector $\mathbf{v}_{t-k}, \ldots, \mathbf{v}_{t-1}$ as input and predicts $\mathbf{v}_t$. Captures non-linear temporal dependencies. Suitable if the dataset is extended beyond 31 days.

**Classifying vibes from artist/track features.** Given a scrobble's artist and track (but *not* its timestamp), can we predict which vibe it belongs to? This asks: *Is there a systematic relationship between what someone listens to and when they listen?*

- **Features**: Artist frequency encoding, track repeat count, album dummy variables, or artist embeddings from SVD.
- **Models**: Multinomial logistic regression (baseline), random forest, gradient-boosted trees (XGBoost/LightGBM).
- **Evaluation**: Multi-class accuracy, macro F1 score, confusion matrix between vibes.
- **Insight**: If accuracy is significantly above random (∼16–20% for 5–6 vibes), this proves that listening content is predictive of listening time — people do have "morning artists" and "night artists."

# 3 Listening Diversity Index (Shannon Entropy)

## 3.1 What It Measures

The **Shannon entropy** of a user's listening distribution measures how *varied* or *uniform* their artist choices are within a given time window. It answers: *Is the user exploring many artists equally, or obsessively replaying one?*

$$H(t) = -\sum_{i=1}^{N_t} p_i(t) \log_2 p_i(t)$$

where $p_i(t)$ is the proportion of plays for artist $i$ in time window $t$, and $N_t$ is the number of distinct artists.

- $H = 0$: The user listened to exactly one artist (zero diversity).
- $H = \log_2 N$: The user listened to all $N$ artists equally (maximum diversity).

## 3.2 Normalized Entropy

To compare across time windows with different numbers of artists, we normalise:

$$H_{\text{norm}}(t) = \frac{H(t)}{\log_2 N_t} \in [0, 1]$$

A value of 0.8 means the user is using 80% of the available diversity.

## 3.3 Time Series Plot

Compute $H(t)$ weekly (or daily) and plot it over the 31-day window. This reveals:

- **Exploration phases**: Periods where $H$ rises sharply — the user is sampling many new artists.
- **Obsession phases**: Periods where $H$ drops — the user found something they love and is playing it on repeat.
- **Weekday/weekend patterns**: Diversity might systematically differ between work days and free days.

## 3.4 Cross with Vibes

Compute entropy *separately for each vibe*. This answers questions like:

- Are Late Night sessions more diverse than Morning sessions?
- Do Weekend Vibes have higher entropy than Weekday Grind?
- Does the user always listen to the same artist at a particular time, but explore at another?

## 3.5 Machine Learning Approach

**Forecasting diversity.** $H(t)$ is a univariate continuous time series — any standard time series model applies.

- **SARIMA**: Directly applicable. Seasonal period of 7 if weekly cycles exist in diversity.
- **SARIMAX with exogenous variables**: Use play count, vibe proportions, and day-of-week as exogenous regressors to predict diversity. This asks: *Can we forecast how varied someone's listening will be based on when and how much they listen?*
- **Random Forest Regressor**: For non-linear relationships. Input features: lagged entropy values, day of week, total plays, number of unique artists in the previous $k$ days.

**Anomaly detection.** Sudden drops in diversity (obsession events) or spikes (exploration bursts) are anomalies. An **Isolation Forest** or **autoencoder** trained on normal diversity patterns could flag unusual listening behaviour.

# 4 New Artist Discovery Rate

## 4.1 What It Measures

For each scrobble, we can check whether the artist has *ever been played before* by that user (up to that point in time). This yields two metrics:

1. **New artists per day/week**: The absolute count of first-ever artist listens within the time window.
2. **Discovery rate**: The fraction of scrobbles in a time window that feature a never-before-heard artist.

## 4.2 Cumulative Discovery Curve

The running total of unique artists over time follows a characteristic shape:

- **Early**: Steep rise (most artists are "new").
- **Later**: Flattening curve (the user has heard most of their rotation).

The *rate of flattening* reveals whether the user's taste is saturating or still actively growing. Over a longer dataset, this curve typically follows **Heaps' law**: $V(n) \approx K \cdot n^{\beta}$ where $V(n)$ is the vocabulary (unique artists) after $n$ plays, and $\beta < 1$.

## 4.3 Time Series Plot

Plot the daily discovery rate as a line chart with a rolling average overlay. Spikes indicate days where the user actively sought out new music; troughs indicate comfort-listening days.

## 4.4 Cross with Vibes

Compute the discovery rate separately per vibe. This answers:

- *Do users discover new music more during Late Night sessions or during Weekend Vibes?*
- *Is the Weekday Grind purely repetitive, or does it also introduce new artists?*

## 4.5 Machine Learning Approach

**Forecasting discovery rate.** This is another univariate time series, but with a particular characteristic: it tends to **decrease over time** as the user accumulates more known artists.

- **SARIMA with trend**: The $d$ parameter in ARIMA handles the decreasing trend. A model of $(1, 1, 1)(1, 0, 1, 7)$ could work.
- **Exponential smoothing (ETS)**: Suitable for data with a decaying trend. A damped trend model (ETS with trend = damped) would naturally capture the flattening behaviour.

- **Poisson regression**: Since "new artists per day" is a count, a Poisson or negative binomial regression with time and vibe features as regressors is statistically appropriate.

**Classification: Will the user discover a new artist today?** Binary: yes/no, at least one new artist. Features: day of week, vibe distribution of previous days, recent entropy, cumulative unique count so far. Model: logistic regression or gradient-boosted classifier.

# 5 Artist Rank Dynamics (Personal Billboard Chart)

## 5.1 What It Measures

For each time window (day or week), rank the user's top $N$ artists by play count. Track how these ranks change over time — like a personal Billboard Hot 100.

## 5.2 Time Series Plot

A **bump chart** (ranked line chart with inverted $y$-axis: rank 1 at top). Each artist is a coloured line. Lines crossing each other show taste evolution.

**What to look for**:

- **Stable monarchs**: Artists that hold rank 1 consistently.
- **Meteoric rises**: An artist suddenly jumping from unranked to top 3 (a new obsession).
- **Fadeouts**: Artists dropping out of the top $N$ over time (novelty wearing off).

## 5.3 Machine Learning Approach

**Rank forecasting.** Predicting an artist's rank next week given their recent trajectory.

- **Ordinal regression**: Ranks are ordinal — standard regression treats the difference between rank 1 and 2 the same as between rank 10 and 11, which is not ideal. Ordinal regression preserves the ordering without assuming equal spacing.
- **Learning-to-rank (LTR)**: Algorithms like **LambdaMART** (used in search engine ranking) can learn to predict artist rankings from features like recent play counts, trajectory direction, and consistency.
- **Per-artist ARIMA**: Fit a separate ARIMA model to each artist's daily play count, then rank the resulting forecasts. This is a simple but effective decomposition.

**Detecting rank change events.** A **change-point detection** algorithm (e.g. PELT, BOCPD) applied to each artist's play count series can identify the exact day a new obsession began or an old favourite faded.

# 6 Listening Session Analysis

## 6.1 Defining a Session

A **listening session** is a continuous period of listening activity. We define a session boundary as a gap of more than 30 minutes between consecutive scrobbles. Every scrobble within a session belongs to the same session ID.

## 6.2 Session-Level Metrics

From each session we can extract:

| Metric | Description |
|---|---|
| Duration (minutes) | Time from first to last scrobble in the session. |
| Track count | Number of songs played in the session. |
| Unique artists | Number of distinct artists in the session. |
| Artist entropy | Shannon diversity within the session. |
| Repeat ratio | Fraction of tracks that are repeats within the session. |

## 6.3 Time Series Plots

Aggregate session-level metrics daily or weekly:

- **Average session duration over time**: Reveals whether the user is engaging in longer or shorter listening blocks.
- **Sessions per day**: Distinguishes between "many short sessions" and "few long sessions" — both could produce the same raw play count but indicate very different behaviour.
- **Average tracks per session**: A decreasing trend might indicate declining engagement; an increasing trend suggests deeper listening.
- **Average unique artists per session**: Measures within-session exploration. A session with 20 tracks from 1 artist is very different from 20 tracks from 15 artists.

## 6.4 Machine Learning Approach

**Forecasting session characteristics.** Each daily aggregate (e.g. mean session duration) is a time series that can be modelled independently.

- **SARIMA / ETS**: For the aggregate metrics (mean duration, session count per day).
- **Multivariate VAR**: Model all session metrics jointly. Session count, mean duration, and mean diversity are likely correlated — a day with many sessions might have shorter average duration.
- **Gaussian Process Regression**: For smooth, uncertainty-aware forecasts of session metrics with limited data (31 days). GPs naturally handle small datasets and provide confidence intervals.

**Session type classification.** Cluster sessions into types: "background listening" (long, low diversity, one artist), "active exploration" (many unique artists, high entropy), "party mode" (Friday night, high energy). Then track how the proportion of session types changes over time — this is itself a compositional time series (similar to vibe proportions).

- **K-Means / Gaussian Mixture Models**: On the feature vector (duration, track count, unique artists, entropy, repeat ratio) to discover natural session types.
- **Hidden Markov Model (HMM)**: Model the user as transitioning between latent "listening states" (background, active, binge). The HMM captures the temporal dynamics of transitions between session types — e.g. a binge session is more likely to follow another binge session than to follow a background session.

# 7 Loyalty Score

## 7.1 What It Measures

The **loyalty score** is the percentage of a user's plays in a time window that go to their all-time top $K$ artists (e.g. $K = 10$):

$$L(t) = \frac{\text{plays of top-}K\text{ artists in window } t}{\text{total plays in window } t} \times 100$$

- $L = 100\%$: The user exclusively listens to their favourites.
- $L \approx 0\%$: The user is entirely outside their usual rotation.

## 7.2 Time Series Plot

Plot $L(t)$ daily with a filled area chart. Overlay a horizontal reference line at 50%. Periods above 50% are "loyal" phases; periods below are "adventurous" phases.

## 7.3 Relationship to Diversity

Loyalty and diversity are **inversely related** but not identical. A user could have low loyalty (rarely playing top-10 artists) but also low diversity (obsessing over one *new* artist). Together, they paint a more complete picture.

## 7.4 Machine Learning Approach

- **SARIMA**: Loyalty is bounded (0–100%) but often stays in the 20–80% range, so SARIMA works well in practice.
- **Beta regression**: Since $L(t)/100 \in (0, 1)$ is a proportion, beta regression is the statistically appropriate model. Predictors: day of week, recent diversity, recent session count.
- **Threshold classification**: Define phases as "loyal" ($L > 60\%$) or "exploratory" ($L \leq 60\%$) and train a classifier to predict tomorrow's phase from recent features. Model: logistic regression, SVM, or a simple decision tree.

# 8 Seasonal and Cyclical Patterns

## 8.1 Weekly Cycles

The most prominent cycle in listening data is the **weekly cycle**. With our data spanning exactly 31 days (approximately 4.4 weeks), we can detect weekly patterns by:

1. **Day-of-week aggregation**: Average play count (and all other metrics) by day of week across all weeks. Visualise as a bar chart.
2. **Fourier analysis**: Compute the periodogram or FFT of the daily time series to identify dominant frequencies. A peak at frequency $f = 1/7$ confirms weekly seasonality.

## 8.2 Artist Seasonality Heatmap

For each of the top $N$ artists, compute total plays per day of week (aggregated across all weeks). Display as a heatmap with artists on the $y$-axis and days of week on the $x$-axis.

**What to look for**:

- Artists that appear exclusively on weekends.
- Artists that spike on specific days (e.g. "Monday motivation" artists).
- Uniform artists that are listened to equally every day (true favourites vs. context-dependent artists).

## 8.3 Hourly Distribution (Listening Clock)

A **radial (polar) bar chart** showing play count by hour of day, plotted around a 24-hour clock. Optionally split into two overlaid rings: weekday vs. weekend. This is not a time series per se, but it *aggregates* the time series into a powerful visual summary.

## 8.4 Machine Learning Approach

**Fourier features for SARIMAX.** Instead of relying solely on SARIMA's seasonal component, we can explicitly construct Fourier features:

$$\sin\left(\frac{2\pi kt}{7}\right), \quad \cos\left(\frac{2\pi kt}{7}\right) \qquad k = 1, 2, 3$$

These sine/cosine pairs, passed as exogenous variables to SARIMAX, capture weekly patterns more flexibly than a single seasonal period.

**Facebook Prophet.** Prophet is designed specifically for time series with strong seasonal effects and multiple seasonality levels. It would model:

- **Weekly seasonality**: Automatic weekday/weekend patterns.
- **Trend**: Piecewise linear trend over the month.
- **Holidays/events**: Could mark specific dates (e.g. New Year's Day falls within our January 2021 data) as special events.

# 9 Track Repeat Intensity

## 9.1 What It Measures

How many times the user replays the *same track* within a day or session. This captures the phenomenon of "having a song on repeat" — a strong signal of emotional connection.

$$R(t) = \max_{\text{track } j} (\text{play count of } j \text{ on day } t)$$

Alternatively, the **average repeat count**: mean number of times each played track is repeated within the day.

## 9.2 Time Series Plot

Plot $R(t)$ daily. Spikes indicate days with obsessive replays. Cross with the identity of the track being repeated to tell a "story" of obsessions over time.

## 9.3 Machine Learning Approach

- **Poisson / Negative Binomial regression**: Repeat counts are non-negative integers with possible overdispersion. A count model with day-of-week and vibe features as regressors is natural.
- **Zero-inflated models**: Many tracks are played exactly once. A zero-inflated Poisson model handles the excess of "no repeats" entries.
- **Survival analysis**: Model the "time until an artist is replayed" using Cox proportional hazards or Kaplan–Meier estimates. Artists with short replay times are "sticky"; artists with long replay times were one-hit curiosities.

# 10 Artist Lifespan Analysis

## 10.1 What It Measures

For each artist, compute:

- **First listen date**: The earliest scrobble of that artist.
- **Last listen date**: The most recent scrobble.
- **Active span**: Last − First (in days).
- **Total plays within span**: How intensely the artist was played during their active period.

## 10.2 Visualisation

A **Gantt-style chart** with artists on the $y$-axis and time on the $x$-axis. Each artist gets a horizontal bar from first to last listen, with colour or thickness encoding play intensity. This shows which artists "stuck" and which were fleeting encounters.

### 10.3 Machine Learning Approach

**Predicting artist longevity.** Given the features of an artist's first few listens (time of day, vibe, number of initial plays, whether it was discovered via a binge session), can we predict whether the artist will still be in rotation a week later?

- **Survival models**: Cox regression with features from the first session. The "event" is the artist being dropped (no plays for $> k$ days).
- **Binary classification**: Will this artist be played again after 7 days? Features: initial play intensity, vibe of discovery, session type, diversity at time of discovery.
- **Random forest / XGBoost**: For the binary classifier, with feature importance analysis revealing what predicts artist stickiness.

## 11 Listening Gap Analysis

### 11.1 What It Measures

Periods with **zero or near-zero scrobbles**: the user stopped listening. These gaps correspond to real-life events — sleep, work, travel, social activities, or simply not being in the mood for music.

### 11.2 Time Series Plot

Overlay "gap regions" (shaded bands) on top of the standard daily play count chart. Alternatively, plot the **daily maximum gap duration** (longest silence in hours) as its own time series.

### 11.3 Machine Learning Approach

- **Anomaly detection**: Unusually long gaps within a period where the user is normally active could flag interesting behavioural changes. Isolation Forest or DBSCAN on gap durations.
- **Predicting active periods**: Given the time of day and day of week, predict whether the user will be listening. This is a binary time series classification problem solvable with logistic regression or an HMM with "active" and "inactive" states.

## 12 Rolling Unique Artist Count

### 12.1 What It Measures

A **rolling window** (e.g. 7 days) count of unique artists. Unlike instantaneous diversity (entropy), this captures the breadth of the user's repertoire over a sustained period. It smooths out daily fluctuations and reveals the underlying trend.

### 12.2 Time Series Plot

A smooth line chart of the 7-day rolling unique artist count. Compare it against the rolling play count to see if the user is listening more *broadly* or just listening *more*.

### 12.3 Machine Learning Approach

- **SARIMA**: Applicable to the smoothed series.
- **Regression**: Predict the rolling unique count from upcoming features (is a holiday approaching? Is it a weekend streak?).

## 13 Multi-User Comparative Analysis

With 11 users in the dataset, every metric above can be computed per user and compared.

### 13.1 Possible Comparisons

| Comparison | Insight |
|---|---|
| Diversity trajectories | Which users explore more over time? Who is most set in their ways? |
| Vibe profiles | Do some users listen exclusively at night while others are morning listeners? |
| Discovery curves | Whose taste is saturating fastest? Who is the most adventurous? |
| Loyalty rankings | Who is the most loyal listener? Who has the most scattered taste? |
| Session patterns | Who listens in long focused sessions vs. many short bursts? |

### 13.2 Machine Learning Approach

**User clustering.** Compute a feature vector for each user: average diversity, average loyalty, dominant vibe, discovery rate, session characteristics. Apply **K-Means** or **hierarchical clustering** to group users into "listener types."

**User-level forecasting.** Fit a **hierarchical (multi-level) time series model** across all users simultaneously. Global patterns (everyone listens less on Mondays) inform individual forecasts, and individual-level parameters capture user-specific behaviour. This approach can be implemented with **mixed-effects models** or **global neural networks** (one LSTM trained on all users with user embeddings).

## 14 Comprehensive Summary of All Possibilities

The following table summarises every metric, its time series potential, and the recommended machine learning approach:

| Metric | Time Series Type | Best Plot | Recommended ML Model |
|---|---|---|---|
| Daily Play Count | Univariate, integer | Line chart | SARIMA (baseline) |
| Vibe Proportions | Multivariate, compositional | Stacked area chart | VAR, Dirichlet regression, LSTM |
| Shannon Entropy (Diversity) | Univariate, continuous | Line with filled area | SARIMAX, Random Forest, GP |
| Discovery Rate | Univariate, decaying | Line + rolling avg | ETS (damped trend), Poisson regression |
| Artist Rankings | Multivariate, ordinal | Bump chart | Ordinal regression, LambdaMART |
| Session Metrics | Multivariate, continuous | 2×2 subplot panel | VAR, Gaussian Process |
| Session Types | Compositional | Stacked bar chart | K-Means + HMM |
| Loyalty Score | Univariate, bounded | Filled line chart | Beta regression, SARIMA |
| Seasonal Patterns | Periodic | Heatmap, polar chart | Prophet, Fourier + SARIMAX |
| Repeat Intensity | Univariate, count | Bar chart with labels | Poisson / NB regression |
| Artist Lifespan | Non-temporal (cross-sectional) | Gantt chart | Survival analysis (Cox) |
| Listening Gaps | Univariate, duration | Shaded overlay | Isolation Forest, HMM |
| Rolling Unique Artists | Univariate, smooth | Smooth line chart | SARIMA, regression |
| User Clusters | Cross-sectional | Scatter / radar chart | K-Means, hierarchical |

# 15 Recommended Priority Order

Given the constraints of the dataset (31 days, 11 users), some analyses are more feasible and impactful than others. We recommend the following priority:

1. **Vibe Composition Over Time** — The most novel and visually compelling analysis. Directly builds on existing feature engineering. Enables all downstream cross-analyses.
2. **Shannon Diversity Index** — Conceptually rich, easy to compute, and produces a beautiful time series that tells a psychological story about each user.
3. **Discovery Rate** — Natural complement to diversity. Together, diversity + discovery fully characterise the "exploration vs. exploitation" behaviour.
4. **Session Analysis** — Adds a dimension no other metric captures: *how* the user engages (deep sessions vs. scattered plays).
5. **Loyalty Score** — Quick to compute, intuitive, and works as a high-level summary of taste stability.

6. **Artist Rank Chart** — Visually distinctive (bump chart), tells individual artist "stories," and is immediately understandable.
7. **Seasonal Heatmap and Listening Clock** — Best for presentation slides. Not a time series per se, but essential for understanding the data.
8. **Track Repeat Intensity and Artist Lifespan** — More specialised but still interesting. Implement if time allows.
9. **Multi-User Comparison** — A natural final chapter, applying all the above across the 11 users to find listener archetypes.

# 16 Constraints and Considerations

## 16.1 Limited Time Span (31 Days)

The dataset covers only January 2021. This limits:

- **Seasonal analysis**: We cannot detect monthly or yearly cycles. Weekly cycles are the longest detectable pattern.
- **Model training**: 31 data points (daily) is small for complex models. SARIMA and simple regression are appropriate; deep learning (LSTM) is not, unless we aggregate at finer granularity (hourly: $31 \times 24 = 744$ points).
- **Forecasting horizon**: With 31 days total and an 80/20 split, we forecast only $\sim$6 days. This is sufficient for demonstration but limits real-world applicability.

## 16.2 Data Granularity

The data has **minute-level resolution** (Date + Time). This enables session analysis and fine-grained vibe assignment but also means aggregation choices matter:

- **Hourly aggregation**: 744 points, good for diversity and session metrics.
- **Daily aggregation**: 31 points, good for SARIMA and high-level trends.
- **Weekly aggregation**: $\sim$4 points, too few for ML but good for summary statistics.

The ideal approach is to compute metrics at **daily resolution** for time series modelling, and at **hourly resolution** when more data points are needed (e.g. for neural models or gap analysis).

## 16.3 Missing Genre Information

The dataset contains Artist, Track, and Album but **no genre or audio features**. This means:

- We cannot directly compute "vibe" in the musical sense (e.g. energy, valence, danceability).
- However, genre can be **approximated** by using artist-level clusters from SVD on the user-artist interaction matrix (as explored in the `svd_kmeans_test/` directory).

- External APIs (Last.fm tags, Spotify audio features) could enrich the dataset if desired.

# 17 Conclusion

Raw play count is merely the surface of what `fm_data.csv` can reveal. By reframing the data through the lens of **vibes**, **diversity**, **discovery**, **sessions**, and **loyalty**, we unlock a rich multi-dimensional portrait of listening behaviour.

The "vibes" framework is the most significant conceptual contribution: it transforms timestamps from raw numbers into meaningful behavioural categories, enabling both human-interpretable summaries and powerful machine learning features. When crossed with metrics like Shannon entropy or discovery rate, vibes reveal *why* a user listens differently at different times — not just *that* they do.

Every metric explored in this document is computable from the existing data, produces a meaningful time series plot, and has a well-defined machine learning modelling strategy. The recommended priority order balances novelty, visual impact, computational feasibility, and insight depth.

The next step is to implement these analyses, starting with the vibe framework and diversity index, building toward a comprehensive dashboard that integrates all perspectives into a unified view of music listening behaviour.